

PARAMJEET ROUT

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Professional Summary

Data Scientist (Gen AI) with 2.8+ years of experience building production-grade **LLM applications, multi-agent RAG systems, and ML-powered fraud detection**. Delivered SurveySenseAI (LangGraph-based multi-agent conversational intelligence for survey research) and an AI Agent-Powered Customer Support Assistant with **Redis-backed** state and **CI/CD**. Implemented **duplicate-user/fraud detection** using phonetic + edit-distance matching and **XGBoost**, achieving 70% precision. Strong foundation in **OLTP/OLAP, ETL pipelines, SQL, and root-cause analysis**, with hands-on experience in Power BI for business analytics.

Experience

Data Scientist (GenAI) — Unimrkt Response LLP

Oct 2025 – Present | Gurugram (Hybrid)

- Architected a LangGraph-first multi-agent RAG system (SurveySenseAI) with two specialized agents: a **Generic FAQ agent** (RAG over survey documentation) and an **Account-Specific Data agent** (secure, user-scoped queries), routed via **LangGraph conditional edges** and shared state for **parallel, autonomous execution**.
- Implemented retrieval over **FAISS vector indexes** with chunking and token-optimization strategies to balance LLM accuracy, latency, and API cost in a live production environment; exposed via FastAPI endpoints with Hugging Face models for generation.
- Designed and maintained **OLTP-to-OLAP data pipelines** in PostgreSQL to support RAG and analytics workloads, ensuring isolation of source transactional DBs from heavy read/query traffic; leveraged **Redis** for caching and session state in the GenAI service.
- Built a **duplicate-user/fraud detection** pipeline combining phonetic hashing (Soundex/Metaphone) and normalized **Levenshtein/Jaro-Winkler** scores with an XGBoost classifier over **identity, geolocation, and behavioral features**; achieved 70% precision with MLflow tracking for model evaluation (precision, recall, F1).
- Containerized the GenAI service with Docker and integrated **Azure Pipelines (azure-pipelines.yml)** for CI/CD, including **JWT-based authentication**, making the system production-ready beyond prototype stage.

Data Scientist Associate — Tech Mahindra

Mar 2024 – Sept 2025 | Ahmedabad

- Built **Python + Azure Functions pipelines** for PDF parsing, metadata extraction, and data cleaning to automate document-processing workflows.
- Implemented **JWT-based, role-aware** document retrieval with FastAPI middleware; integrated OpenAI, Gemini, and Together AI behind a unified abstraction layer for model fallback and cost-optimized inference.
- Developed internal **Streamlit + MLflow dashboards** for usage analytics, retrieval accuracy, and latency monitoring; partnered with business teams to convert manual workflows into AI-assisted pipelines.
- Built **Power BI dashboards** and designed **SQL extraction** queries and optimized data models to support multiple concurrent reporting pipelines, reducing manual reporting effort by 60%.

Leadership and Open-Source Contributions – OSCG'26

- Supervised contributors; reviewed and merged 30+ PRs across AI and automation-based projects.
- Mentored multiple teams in implementing clean, maintainable code and robust AI-driven systems.

Projects

SurveySenseAI : Multi-Agent Conversational Intelligence | github.com/routparam12/SurveySense-AI-Multi-agent-conversational-Intelligence-for-Survey-Research

Oct 2025 – Dec 2025

LangGraph · LangChain · FastAPI · Hugging Face · FAISS · Redis · PostgreSQL · Docker · Azure Pipelines

- Architected a LangGraph-first multi-agent RAG system with two specialized agents: a **Generic FAQ agent** (RAG over survey docs) and a specific **account-data agent** (secure, user-scoped queries), routed via LangGraph conditional edges and shared state for parallel, autonomous execution.
- Designed agent routing and coordination layer** using LangGraph conditional edges and **state management** to classify the correct agent at runtime, enabling parallel multi-agent execution.
- Designed the full RAG pipeline: **OpenAI text-embedding-3-large (1024 dimensions)**, **10–20%** chunk size with **10–12%** overlap, **ChromaDB vector store**, and **top-k** semantic search; unified LLM layer over OpenAI GPT + Google Gemini with fallback.

- Implemented 9-type **persistent memory** (SHORT TERM, LONG TERM, ENTITY, FACT, PREFERENCE, PROCEDURAL, EPISODIC, SEMANTIC, WORKING) in **PostgreSQL + SQLAlchemy**, **Redis session caching**; deployed with **Docker + Azure CI/CD**.

FairExam.in : AI Exam Engine for Leak-Proof Assessments | github.com/routparam12/FairExam-AI-Exam-Engine-for-Leak-Proof-Assessments Apr 2026 – Ongoing

Next.js · FastAPI · LangChain · LangGraph · RBAC · PyTorch · ChromaDB · Firebase · Supabase · AWS

- **Built and deployed a full-stack AI exam platform** at fairexam.in — **Next.js** frontend, **FastAPI** backend hosted on **AWS**, **Firebase** Authentication (Google OAuth) for seamless login, and **Supabase PostgreSQL** for persistent storage.
- **Eliminated paper leaks structurally** — MCQs generated dynamically at exam time via **LlamaIndex + ChromaDB** RAG pipeline and **PyTorch**-backed embedding models; questions are never stored in any database, making pre-exam leaks structurally impossible.
- **JWT-secured FastAPI backend** with role-based access control — admin-only syllabus ingestion (PDF/DOCX), LangGraph-orchestrated agentic AI exam flow, paginated 20-question delivery with shuffle logic; architected for high-concurrency public service examinations.
- Applied **token optimization and model evaluation** — context window budget management, chunk sizing, and difficulty normalization metrics ensuring consistent question quality across all candidates.

Fraud User Detection System | github.com/routparam12/SimilarAccountDetection.ML

Feb 2026 – Apr 2026

Python · Fuzzy Logic · Decision Tree · MLflow · Streamlit · Azure deployment

- **Designed and deployed a duplicate-user/fraud detection pipeline fusing rule-based entity resolution with supervised learning**: phonetic hashing (Soundex/Metaphone) and normalized Levenshtein/Jaro–Winkler scores for fuzzy matching, combined with an **XGBoost classifier** to score high-risk and duplicate accounts.
- Engineered features across identity, network, and behavior layers: **email/token similarity metrics, IP geolocation-derived risk signals, device/browser fingerprints, and session-level behavioral patterns**; handled class imbalance via threshold tuning.
- **Achieved 70% precision** at operating point (7/10 flagged accounts confirmed fraudulent); implemented experiment tracking with MLflow and a Streamlit monitoring dashboard for drift and feature importance analysis.

Technical Skills

Generative AI & LLM: LangChain, LangGraph, LlamaIndex, Hugging Face, RAG Pipelines, Agentic AI, Multi-Agent Systems, Prompt Engineering, OpenAI GPT, Google Gemini API, Token Optimization, Model Fallback

Vector & Search: FAISS, ChromaDB, Elasticsearch, Pinecone, Vector Database, Semantic Search, Embedding Services

ML & NLP: scikit-learn, Decision Tree, Fuzzy Matching, Feature Engineering, Model Evaluation (Precision/Recall/F1), MLflow, Natural Language Processing (NLP)

Backend & Data: FastAPI, Python, PostgreSQL, SQLAlchemy, Redis, Docker, Azure Pipelines, Git/GitHub, Streamlit

Data Engineering: OLTP, OLAP, ETL Pipelines, SQL Optimization, Logstash, Data Warehouse, Power BI, DAX

Certifications

- **Anthropic Academy** — Building with the Claude API (2026)
- **Microsoft PL-300** Power BI Data Analyst — *In Progress (2026)*
- **Microsoft DP-900** Azure Data Fundamentals — *In Progress (2026)*
- Google Antigravity Program — Generative AI & Applied AI Concepts (2025)
- RAG Learning & Bot Building — Hands-on Completion (2025)

Education

B.Tech — Computer Science & Engineering — **Gandhi Institute of Excellent Technocrats (GIET), Bhubaneswar**

2020 – 2024 | CGPA: 8.4/10